Lecture 03

Your python project

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Lecture dates

Lecture	Date	T.H.A.	Room
1: Kickoff	17/2	N	RZ D8
2: AI assistance and tools	24/2	N	RZ D8
3: Project-based teamwork	3/3	N	RZ D8
4: Intro to Git	10/3	N	RZ D8
5: Git continued	17/3	N	TBA
6: Git advanced	24/3	Y	RZ D8
7: AI-assisted Python	31/3	Y	TBA
8: The 3 R's in Python	14/4	Y	TBA
9: Python packaging ecosystem	21/4	N	RZ D8
10: OOP	28/4	Y	RZ D8
11: Testing and debugging	5/5	Y	TBA
12: Joker lecture	12/5	N	RZ D8
13: Project distribution	19/5	N	RZ D8
14: Project marketplace	26/5	N	RZ D8

T.H.A: Take home assignment

TBA: Lecture room to be announced

Today's learning objectives

You spent the first hour today getting to know your team. Next, we're going to tackle how your project might look like!

1. Explore project examples
2. Brainstorm project idea
3. Project proposals & voting
4. Iterate with your team

Some world-class examples

PyTorch

GitHub

PyTorch is a powerful machine learning framework primarily used for deep learning applications, offering a flexible "dynamic computational graph" that is highly favored by researchers and AI developers.

SciPy

GitHub

SciPy is a fundamental library for scientific computing that builds on NumPy to provide efficient routines for numerical integration, optimization, signal processing, and statistics.

PyVista

GitHub

PyVista serves as a streamlined interface for the Visualization Toolkit (VTK), making it much easier to create, manipulate, and plot complex 3D meshes and spatial datasets.

Project communities

- project-based-learning
- Project-Ideas-And-Resources
- Python-for-Scientists
- awesome python
- f****ng-awesome-python
- python-mini-project

Geophysical Python Ecosystem

ObsPy

GitHub

ObsPy is a toolbox for seismology that provides unified access to seismic data formats and web services, facilitating signal processing and data analysis for earthquake monitoring.

SimPEG

GitHub

SimPEG is an open-source framework for simulation and gradient-based parameter estimation in geophysics, supporting diverse methods like electromagnetics and gravity.

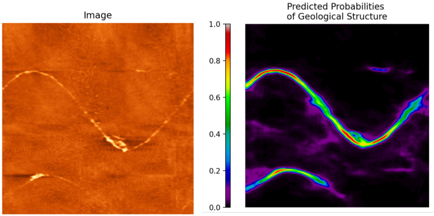
GemPy

GitHub

GemPy is an open-source library for implicit 3D structural geological modeling, capable of constructing complex models of folded and faulted layers from sparse data.

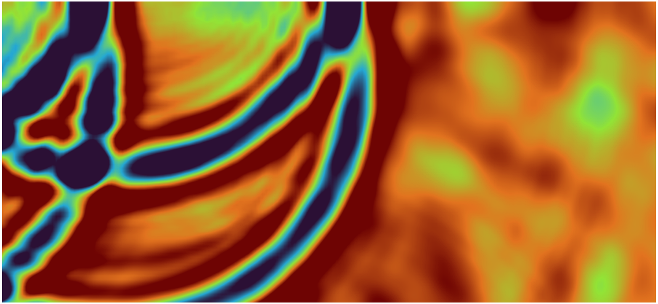
DeepLogger

Create a graphical interface where users can interact with borehole geophysical images, either for labeling or segmentation through machine learning models. Can train models for predicting geological structures. Guided by me and Marian Hertrich, Bedretto Lab / Mathematical Geophysics.



Interactive PDEs

The project will develop an interactive Python-based simulator for fundamental partial differential equations (PDEs) commonly encountered in physics and engineering. The tool will allow users to visualize and interact with numerical solutions of selected PDEs in real time. Guided by Arnaud Mercier and Cedric Schmelzbach, Mathematical Geophysics.



Moon Landing Site

The goal is to provide a tool that will evaluate a landing site on the Moon and create hazard maps that will guide robotic path planning from a landing site to an exploration target. This includes getting data from a given region of interest, determining slope, rock distribution, crater distribution, . . . and come up with a recommendation for a landing site, a hazard map and path planning. Guided by Anna Mittelholz, Planetary Sciences.